Visual Instruction Tuning

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Abstract

Instruction tuning large language models (LLMs) using machine-generated instruction-following data has been shown to improve zero-shot capabilities on new tasks, but the idea is **less explored** in the multimodal field.(开篇点明任务背景:指令微调可以提升LLM零样本学习能力,但在多模态领域尚未得到充分探索) We present the **first attempt** to use language-only GPT-4 to generate multimodal language-image instruction-following data.(承接less explored,延伸创新点) By instruction tuning on such generated data, we introduce LLaVA: **L**arge **L**anguage **a**nd **V**ision **A**ssistant, an end-to-end trained large multimodal model that connects a vision encoder and an LLM for general-prupose visual and language understanding.(❶提出核心方法) To facilitate future research on visual instruction following, we construct two evaluation benchmarks with diverse and challenging application-oriented tasks.(❷构建评测bench) Our experiments show that LLaVA demonstrates impressive multimodal chat abilities, sometimes exhibiting the behaviors of multimodal GPT-4 on unseen images/instructions, and yeilds a 85.1% relative score compared with GPT-4 on a synthetic multimodal instruction-following dataset.(❸总结实验效果) When fine-tuned on Science QA, the synergy of LLaVA and GPT-4 achieves a new state-of-the-art accuracy of 92.53%.(❹易用性与泛化性证明) We make GPT-4 generated visual instruction data, our model, and code publicly avaliable.

1 Introduction

Humans interact with the world through many channels such as vision and language, **as** each individual channel has a unique advantage in representing and communicating certain concepts, and thus facilitates a better understanding of the world.(通过人类的多感官理解开篇,阐述多模态信息的重要性) One of the core aspirations in artificial intelligence is to develop a general-purpose assistant that can effectively follow multi-model vision-and-language instructions, aligned with human intent to complete various real-world tasks in the wild.(进而引出核心目标:让AI助手和人类一样学会理解视觉-语言指令,完成真实环境中的实际任务)

To this end, the community has witnessed an emergent interest in developing language-augmented foundation vision models, with strong capabilities on open-world visual understanding such as classification, detection, segmentation, and captioning, as well as visual generation and editing. We refer readers to the *Computer Vision in the Wild* reading list for a more up-to-date literature compilation.(承接第一段最后一句,写出学术界当前关注的热点方向) In this line of work, each task is solved independently by one single large vision model, with the task instruction implicitly considered in the model design.(当前方向上的任务范式) Further, language is only utilized to describe the image content.(Limitation ❶) While this allows language to play an important role in mapping visual signals to language semantics — a common channel for human communication, it leads to models that usually have a fixed interface with interactivity and adaptability to the user's instruction.(进一步对 Limitation❶ 进行阐述:视觉到语言的映射导致模型具有固定的interface,难以适应用户指令)

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Large Language Models (LLMs), on the other hand, have shown that language can play a wider role: a universal interface for a language-purpose assistant, where various task instructions can be explicitly represented in language and guide the end-to-end trained neural assistant to switch to the task of interest to solve it.(进一步阐述语言在LLM应用中更广泛的作用) For example, the recent success of ChatGPT and GPT-4 have demonstrated the power of aligned LLMs in following human instructions, and have stimulated tremendous interest in developing open-source LLMs.Among them, LLaMA is an open source LLM that matches the performance of GPT-3.(使用开源、闭源模型,综合举例论证instruction-following的重要性) Alpaca, Vicuna, GPT-4-LLM utilize various machine-generated high quality instruction-following samples to improve the LLM's alignment ability, reporting impressive performance compared with proprietary LLMs.(与专用LLM对比,证明instruction-following方法的泛化性与普适应) Importantly, this line of work is *text-only*. (Limitation ❷)

In this paper, we present visual instruction tuning, the **first attempt** to **extend instruction-tuning to the language-image multimodal space**, to pave the way towards building a general-purpose visual assistant.(总结核心方法与贡献) In patticular, our paper makes the following contributions:

- *Multimodal instruction-following data*. One key challenge is the lack of vision-language instruction-following data. We present a data reformation perspective and pipeline to convert image-text pairs into an appropriate instruction-following format, using ChatGPT/GPT-4.(使用ChatGPT/GPT-4将"图片-文本"对转换为指令跟随格式)

- *Large multimodal models*. We develop a large multimodal model (LMM), by connecting the open-set visual encoder of CLIP with the language decoder Vicuna, and fine-tuning end-to-end on our generated instructional vision-language data.(将CLIP视觉编码器与Vicuna语言模型解码器进行桥接并训练) Our empirical study validates the effectiveness of using generated-purpose instruction-following visual agent. When ensembled with GPT-4, our approach achieves SOTA on the Science QA multimodal reasoning dataset.

- *Multimodal instruction-following benchmark*. We present LLaVA-Bench with two challenging benchmarks, with a diverse selection of paired images, instructions and detailed annotations.(提出了指令跟随bechmark)

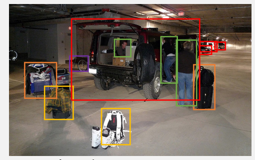
- *Opne-source*. We release the following assets to the public: the generated multimodal instruction data, the codebase, the model checkpoints, and a visual chat demo.

2 Related Work

Multimodal Instruction-following Agents. In computer vision, existing works that build instruction-following agents can be broadly categorized into two classes: (i) End-to-end trained models, which are **separately explored** for each specific research topic.(不同任务独立训练专用模型) For example, the vision-language navigation and take a sequence of actions to complete goals in visual environments. In the image editing domain, given an input image and a written instruction that tells the agents what to do, InstructPix2Pix edits images by following the human instructions. (ii) A system that coordinates various models via LangChain/LLM, such as Visual ChatGPT, X-GPT, MM-REACT, VisProg, and ViperGPT.(通过封装Tookit协调不同模型完成多项不同任务) While sharing the same goal in building instruction-following agents, we focus on developing an end-to-end trained language-vision multimodal model for *multiple* tasks.

Instruction Tuning. In the natural language processing (NLP) community, to enable LLMs such as GPT-3, T5, PaLM and OPT to follow natural language instructions and complete real-world tasks, researchers have explored methods for LLM instruction-tuning, leading to instruction-tuned counterparts such as InstructGPT/ChatGPT, FLAN-T5, FLAN-PaLM, and OPT-IML, respectively.(总起概括对指令微调任务的成果进展) It turns out that this simple approach can effectively improve the zero and few-shot generalization abilities of LLMs.(指令微调简单且高效的作用) It is thus matural to borrow the idea **from NLP to computer vision**. More broadly, the teacher-student distillation ideas with foundation models have been studied in other topics such as image classification.(指令微调跨任务的可行性论证) Flamingo can be viewed as the GPT-3 moment in the multimodal domain, due to its strong performance on zero-shot task transfer and in-context-learning. Other LMMs trained on image-text pairs include BLIP-2, FROMAGe, and KOSMOS-1. PaLM-E is an LMM for enbodied AI.(梳理了系列multimodal LLMs) Based on the recent "best" open-source

Context type 1: Captions
A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes
person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation
Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>
Response type 2: detailed description
The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>
Response type 3: complex reasoning
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

Table 1: One example to illustrate the instruction-following data. The top block shows the contexts such as captions and boxes used to prompt GPT, and the bottom block shows the three types of responses. Note that the visual image is not used to prompt GPT, we only show it here as a reference.

LLM LLaMA, OpenFlamingo and LLaMA-Adapter are open-source multimodal LLMs. While these models present promising task transfer generalization performance in multimodal tasks usually falls short compared to language-only tasks.(总结当下multimodal LLMs局限与不足) In this paper, we aim to fill this gap and study its effectiveness.(gap:模型在多模态任务中泛化性不足,表现不如纯语言任务) Finally, note that visual instruction tuning is different from visual prompt tuning: the former aims to improve the model's **instruction-following abilities**, while the latter aims to improve the **parameter-efficiency** in model adaptation.(延伸总结出视觉指令微调与视觉提示微调的区别)

3 GPT-assisted Visual Instruction data Generation

(论文 Table 1把 captions 和 boxes 一起展示,给读者的印象是两类上下文都用了;但从仓库里实际开源的 few-shot 样本看,conversation 类只用了 captions) The community has witnessed a surge in the amount of public multimodal data such as image-text pairs, ranging from CC to LAION. However, when it comes to multimodal instruction-following data, the available amount is limited, partially because the process for creating such data is time-consuming and less well-defined when human crowd-scouring is considered.(多模态指令跟随数据集不足) Inspired by the success of recent GPT models in text-annotation tasks, we propose to leverage ChatGPT/GPT-4 for multimodal instruction-following data collection, based on the widely existing image-pair data.

For an image $\mathbf{X}_\mathtt{v}$ and its associated caption $\mathbf{X}_\mathtt{c}$, it is natural to create a set of questions $\mathbf{X}_\mathtt{q}$ with the intent to instruct the assistant to describe the image content. We prompt GPT-4 to curate such a list of questions (see details in Appendix). Therefore, a simple way to expand an image-text pair to its instruction-following version is $\mathtt{Human}: \mathbf{X}_\mathtt{q}\ \mathbf{X}_\mathtt{v}\texttt{<STOP>}\ \mathtt{Assistant}: \mathbf{X}_\mathtt{c}\texttt{<STOP>}$. Though cheap to construct, this simple expanded version lacks diversity and in-depth reasoning in both the instructions and responses.(基于上述提示词构建出来的指令跟随模版缺乏多样性和深度推理)

To mitigate this issue, we leverage language-only GPT-4 or ChatGPT as the strong teacher (both accept only text as input), to create instruction-following data involving visual content. Specifically, in order to encode an image into its visual features to prompt a text-only GPT, we use two types of symbolic representations: (i) *Captions* typically describe the visual scene from various perspectives; (ii) *Bounding boxes* usually localize the objects in the scene, and each box encodes the object concept and its spatial location. One example is shown in the top block of Table 14.(在指令中加入Captions和Bboxes,用来缓解多样性和深度性缺失问题)

This symbolic representation allows us encode the image as an LLM-recognizable sequence. We use COCO images and generate three types of instruction-following data. One example per type is shown in the bottom block of Table 14. For each type, we first manually design a few examples. They are the only human annotations we have during data collection, and are used as seed examples in in-context-learning to query GPT-4.

- *Conversation*. We design a conversation between the assistant and a person asking questions about this photo. The answers are in a tone as if the assistant is seeing the image and answering the question. A diverse set of questions are asked about the visual content of the image, including the object types, counting the objects, object actions, object locations, relative positions between objects. Only questions that have definite answers are considered. Please see Appendix for the detailed prompt.(User基于photo向Assistant提问并作答 - QA Piars)
- *Detailed description*. To include a rich and comprehensive description for an image, we create a list of questions with such an intent. We prompt GPT-4 then curate the list (see detailed prompts and curation process in Appendix). For each image, we randomly sample one question from the list to ask GPT-4 to generate the detailed description.(从question list中随机抽取一个问题,使用GPT-4生成一段详细的描述)
- *Complex reasoning*. The above two types focus on the visual content itself, based on which we further create in-depth reasoning questions. The answers typically require a step-by-step reasoning process by following rigorous logic.(在视觉内容基础上,设计推理题目,使模型遵循严谨逻辑推导结果 - QA pairs)

4 Visual Instruction Tuning

4.1 Architecture

The primary goal is to effectively leverage the capabilities of both the pre-trained LLM and visual model. The network archtecture is illustrated in Figure 1. We choose Vicuna as our LLM $f_\phi(\cdot)$ parameterized by ϕ, as it has the best instruction following capabilities in language tasks among publicly available checkpoints.(Vicuna作为LLM解码器并基于ϕ初始化权重)

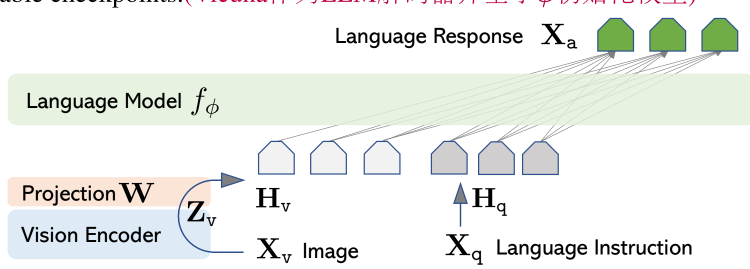


Figure 1: LLaVA network architecture.

For an input image $\mathbf{X}_v$, we consider the pre-trained CLIP visual encoder ViT-L/14, which provides the visual feature $\mathbf{Z}_v = g(\mathbf{X}_v)$. The grid features before and after the last Transformer layer are considered in our experiments.(考虑了最后一层 Transformer 的前后grid features) We consider **a simple linear layer**(亮点) to connect image features into the word embedding space. Specifically, we apply a trainable projection matrix $\mathbf{W}$ to convert $\mathbf{Z}_v$ into language embedding tokens $\mathbf{H}_v$, which have the same dimensionality as the word embedding space in the language model:

$$\mathbf{H}_v = \mathbf{W} \cdot \mathbf{Z}_v, \text{ with } \mathbf{Z}_v = g(\mathbf{X}_v) \tag{1}$$

($\mathbf{W}$为nn.Linear(1024,4096)) Thus, we have a sequence of visual tokens $\mathbf{H}_v$. Note that our simple projection scheme is lightweight, which allows us to iterate data centric experiments quickly.(simple projection的优势) More sophisticated schemes to connect the image and language representations can also be considered, such as gated cross-attention in Flamingo and Q-former in BLIP-2. We leave exploring possibly more effective and sophisticated architecture designs for LLaVA as future work.(使用Flamingo中的门控交叉注意力或BLIP-2中的Q-former,在未来探索针对LLaVA更高效的架构设计)

$\mathbf{X}_{\texttt{system-message}}$ <STOP>
Human : $\mathbf{X}^1_{\texttt{instruct}}$ <STOP> Assistant: $\mathbf{X}^1_{\texttt{a}}$ <STOP>
Human : $\mathbf{X}^2_{\texttt{instruct}}$ <STOP> Assistant: $\mathbf{X}^2_{\texttt{a}}$ <STOP> $\cdots$

Table 2: The input sequence used to train the model. Only two conversation turns are illustrated here; in practice, the number of turns varies based on the instruction-following data. In our current implementation, we follow Vicuna-v0 to set the system message $\mathbf{X}_{\texttt{system-message}}$ and we set <STOP> = ###. The model is trained to predict the assistant answers and where to stop, and thus only green sequence/tokens are used to compute the loss in the auto-regressive model.

4.2 Training

For each image $\mathbf{X}_{\texttt{v}}$, we generate multi-turn conversation data $(\mathbf{X}^1_{\texttt{q}}, \mathbf{X}^1_{\texttt{a}}, \cdots, \mathbf{X}^T_{\texttt{q}}, \mathbf{X}^T_{\texttt{a}})$, where T is the total number of turns. We organize them as a sequence, by treating all answers as the assistant's response, and the instruction $\mathbf{X}^t_{\texttt{instruct}}$ at the t-th turn as:

$$\mathbf{X}^t_{\texttt{instruct}} = \begin{cases} \text{Randomly choose } [\mathbf{X}^1_{\texttt{q}}, \mathbf{X}_{\texttt{v}}] \text{ or } [\mathbf{X}_{\texttt{v}}, \mathbf{X}^1_{\texttt{q}}], & \text{the first turn } t = 1 \\ \mathbf{X}^t_{\texttt{q}}, & \text{the remaining turns } t > 1 \end{cases} \tag{2}$$

This leads to the unified format for the multimodal instruction-following sequence illustrated in Table 2. We perform instruction-tuning of the LLM on the prediction tokens, using its original auto-regressive training objective.(使用原始自回归训练目标对预测Tokens进行指令微调)

Specifically, for a sequence of length L, we compute the probability of the target answers $\mathbf{X}_{\texttt{a}}$ by:

$$p(\mathbf{X}_{\texttt{a}}|\mathbf{X}_{\texttt{v}}, \mathbf{X}_{\texttt{instruct}}) = \prod_{i=1}^{L} p_{\boldsymbol{\theta}}(\boldsymbol{x}_i|\mathbf{X}_{\texttt{v}}, \mathbf{X}_{\texttt{instruct},<i}, \mathbf{X}_{\texttt{a},<i}), \tag{3}$$

(p_θ 里的 θ 就是整个模型里需要被优化器更新的权重集,两阶段训练把它从 $\mathbf{W}$ 逐步扩大到 $\mathbf{W}, \phi$) where $\boldsymbol{\theta}$ is the trainable parameters, $\mathbf{X}_{\texttt{instruct},<i}$ and $\mathbf{X}_{\texttt{a},<i}$ are the instruction and answer tokens in all turns before the current prediction token $\boldsymbol{x}_i$, respectively. Please see Table 2 for an illustration of the prediction tokens. For the conditionals in (3), we explicitly add $\mathbf{X}_{\texttt{v}}$ to emphasize the fact that the image is grounded for all answers, and we omit $\mathbf{X}_{\texttt{system-message}}$ and all previous <STOP> for better readability. For LLaVA model training, we consider a two-stage instruction-tuning procedure.(LLaVA训练涉及两阶段的指令微调过程)

Stage 1: Pre-training for Feature Alignment. To strike a balance between concept coverage and training efficiency, we filter CC3M to 595K image-text pairs. Please see Appendix for details of the filtering process. These pairs are converted to the instruction-following data using the naive expansion method describe in Section 3. Each sample can be treated as a single-turn conversation. To construct the input $\mathbf{X}_{\texttt{instruct}}$ in (2), for an image $\mathbf{X}_{\texttt{v}}$, a question $\mathbf{X}_{\texttt{q}}$ is randomly sampled, which is a language instruction to request the assistant to describe the image briefly. The ground-truth prediction answer $\mathbf{X}_{\texttt{a}}$ is the original caption. In training, we keep both the visual encoder and LLM weights frozen, and maximize the likelihood of (3) with trainable parameters $\boldsymbol{\theta} = \mathbf{W}$ (the projection matrix) only. In this way, the image features $\mathbf{H}_{\texttt{v}}$ can be aligned with the pre-trained LLM word embedding. This stage can be understood as training a compatible visual tokenizer for the frozen LLM.(Stage1:冻结LLM和CLIP,只训练中间的投影层完成视觉空间到语义空间对齐)

Stage 2: Fine-tuning End-to-End. We always keep the visual encoder weights frozen, and continue to update both the pre-trained weights of the projection layer and LLM in LLaVA; i.e., the trainable parameters are $\boldsymbol{\theta} = \{\mathbf{W}, \phi\}$ in (3).(Stage2:冻结CLIP,训练投影层与LLM解码器进行端到端微调) We consider two specific use case scenarios:

- *Multimodal Chatbot*. We develop a Chatbot by fine-tuning on the 158K language-image instruction-following data in Section 3. Among the three types of responses, conversation is multi-turn while the other two are single-turn. They are uniformly sampled in training.
- *Science QA*. We study our method on the ScienceQA benchmark, the first large-scale multimodal science question dataset that annotates the answers with detailed lectures and explanations. Each question is provided a context in the form of natural language or an image. The assistant

Visual input example, Extreme Ironing:	
	
	Source: `https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg`
User	What is unusual about this image?
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.
	[*Start a new conversation, and clear the history*]
User	What's happening in the scene?
LLaVA	The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car. The city street around him is bustling with activity, adding to the unique nature of the scene.
User	What is unusual about this image?
GPT-4	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
User	What is unusual about this image?
BLIP-2	a man is sitting on the back of a yellow cab
User	What is unusual about this image?
OpenFlamingo	The man is drying his clothes on the hood of his car.

Table 3: Example prompt from GPT-4 paper to compare visual reasoning and chat capabilities. Compared to BLIP-2 and OpenFlamingo, LLaVA accurately follows the user's instructions, instead of simply describing the scene. LLaVA offers a more comprehensive response than GPT-4. Even when merely asked to describe the image, LLaVA identifies atypical aspects of the image.

provides the reasoning process in natural language and selects the answer among multiple choices. For training in (2), we organize the data as a single turn conversation, the question & context as $\mathbf{X}_{\texttt{instruct}}$, and reasoning & answer as $\mathbf{X}_{\texttt{a}}$.

"role": "assistant"(豆包输出实例)
Question:
Where is the black SUV parked?
======
Answer:
The black SUV is parked in an underground public garage.
======
Question:
What are the people beside the vehicle doing?
======
Answer:
The group of people are standing outside the SUV and trying to fit all their various pieces of luggage into the vehicle as they get packed for a trip.
======
Question:
Is there luggage around the vehicle?
======
Answer:
Yes, lots of luggage surrounds the SUV, and the people are working to load these belongings into the vehicle.
======
Question:
What advantages does an SUV have for a trip that requires carrying large amounts of luggage?
======
Answer:
SUVs are well-suited for trips with plenty of luggage for several practical reasons.
First, SUVs offer generous interior and rear cargo volume. The rear storage area can hold suitcases, bags and travel gear. Many models allow the back seats to fold flat, which greatly expands available space when people need to pack a large quantity of luggage, just like the scene shown where a group is trying to fit all their belongings into the car.
Second, SUVs often have higher load-bearing capacity. They can handle heavier combined weight of passengers and travel baggage better than many compact sedans. This makes them ideal for group travel with multiple people each bringing their own suitcases.
Third, many SUVs have a higher roofline and taller rear opening. It is easier to place bulky or large-sized luggage into the trunk, without struggling against a low narrow trunk opening typical of sedans.
In addition, for road trips, most SUVs are designed with comfortable passenger space for several adults. So groups can travel together while also bringing along all their travel necessities. That matches exactly the situation in this garage: a group of people gathering around their vehicle, preparing to load up and start their journey.

Table 4: 豆包输出实例

"role": "user"(5 captions)
A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area.
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip.
Some people with luggage near a van that is transporting it.

Table 5: Conversation Prompt 3

A Instruction-following Prompt

A.1 Conversation Prompt

提示词的结构为: system(任务+约束) → 2×(user 输入示范 → assistant 输出示范) → user(真实输入)。system message 里从头到尾没写"请用 Question: 开头、用 === 分隔",这个输出格式完全由两个 _conv.txt 示范，GPT-4 通过 in-context learning 学会并延续。GPT-4 输出的是纯文本,后续脚本按 === 切成 Q/A 对，再转成训练用的 conversations JSON。

使用Table 6 - Table 7作为zero-shot,Table 5作为真实输入样例构建提示词输入到豆包中得到输出如Table 4所示。

"role": "system"
You are an AI visual assistant, and you are seeing a single image. What you see are provided with five sentences, describing the same image you are looking at. Answer all questions as you are seeing the image.
Design a conversation between you and a person asking about this photo. The answers should be in a tone that a visual AI assistant is seeing the image and answering the question.
Ask diverse questions and give corresponding answers.
Include questions asking about the visual content of the image, including the object types, counting the objects, object actions, object locations, relative positions between objects, etc. Only include questions that have definite answers:
(1) one can see the content in the image that the question asks about and can answer confidently;
(2) one can determine confidently from the image that it is not in the image.
Do not ask any question that cannot be answered confidently.
Also include complex questions that are relevant to the content in the image, for example, asking about background knowledge of the objects in the image, asking to discuss about events happening in the image, etc. Again, do not ask about uncertain details.
Provide detailed answers when answering complex questions. For example, give detailed examples or reasoning steps to make the content more convincing and well-organized. You can include multiple paragraphs if necessary.

"role": "user"(5 captions)
There is a movie theater that displays the show times above the doors.
A red fire hydrant is deep in the snow.
The fire hydrant is in the snow near a recently plowed sidewalk.
This city has had a very hard winter with snow.
A hotel for dogs in the snow in winter.
"role": "assistant"(few-shot样例1)
Question:
What color is the fire hydrant in the image?
======
Answer:
The fire hydrant in the image is red.
======
Question:
Is the fire hydrant fully covered by snow or partially visible?
======
Answer:
The fire hydrant is partially visible, with a significant portion of it covered by the deep snow.
======
Question:
What is the condition of the sidewalk near the fire hydrant?
======
Answer:
The sidewalk near the fire hydrant has been recently plowed, so it is relatively clear of snow compared to the surrounding area.
======
Question:
What does the image suggest about the winter season in the city?
======
Answer:
The image suggests that the winter season in the city has been quite harsh and challenging. The significant snow accumulation seen in the image indicates that the city has experienced heavy snowfall, which can lead to various difficulties and disruptions for its residents.
======
For instance, the deep snow covering the fire hydrant could potentially hinder its accessibility in case of emergencies, which poses safety concerns. The recently plowed sidewalk implies that the city's maintenance crews have been actively working to keep the walkways clear and safe for pedestrians, but the sheer amount of snowfall might make it difficult to keep up with the continuous snow removal efforts.
Furthermore, such extreme winter conditions can impact transportation, leading to traffic issues, delays in public transit, and increased risks of accidents due to slippery roads. It can also cause problems for businesses and schools, as people might face challenges commuting to work or attending classes. Additionally, the heavy snow can put extra strain on infrastructure, such as roofs and power lines, increasing the likelihood of structural damage or power outages.
In conclusion, the image of the red fire hydrant deep in the snow and the recently plowed sidewalk suggest that the city has faced a particularly severe winter season, with substantial snowfall that has likely caused various challenges and disruptions for its residents and infrastructure.

Table 6: Conversation Prompt 1

"role": "user"(5 captions)
A man is skiing in the open snow covered hills.
A skier is making his way into the snow.
A skier on the foothills of a huge mountain range.
A skier looks at mountains while standing near a trail sign.
A single person skiing by an area with a lot of bushes.
"role": "assistant"(few-shot样例2)
Question:
What is the skier doing?
======
Answer:
The skier is skiing in the snow-covered hills, making his way through the snow and enjoying the beautiful mountain scenery. They are also standing near a trail sign, which indicates their location on the mountain.
======
Question:
Are there any other people or skiers visible in the image?
======
Answer:
No, there is only a single person skiing in the image, and they are surrounded by an area with a lot of bushes.
======
Question:
What kind of skiing is the person doing? Cross-country or downhill?
======
Answer:
Based on what I can see in the photo, it is not possible to definitively determine whether the skier is doing cross-country or downhill skiing. However, given the presence of a trail sign and the fact that they are skiing in the foothills, it is more likely that the skier is engaged in cross-country skiing. Downhill skiing typically takes place on more steep slopes and involves ski lifts, whereas cross-country skiing often occurs on relatively flatter terrain with marked trails.
======
Question:
Can you explain cross-country and downhill skiing a bit more?
======
Answer:
Cross-country skiing and downhill skiing are two distinct types of skiing, each with its own techniques, equipment, and terrain preferences.
======
1. Cross-Country Skiing:
Also known as Nordic skiing, cross-country skiing involves traversing relatively flat terrain or gentle slopes using skis and poles. It is a popular recreational and competitive sport that focuses on endurance, balance, and aerobic fitness. Cross-country skiers use their own locomotion to move across snow-covered terrain, rather than relying on ski lifts or other forms of assistance. There are two main techniques in cross-country skiing:
- Classic: This technique involves skiing in parallel tracks, using a forward-and-backward gliding motion while keeping the skis parallel to each other. The skier uses their poles to propel themselves forward.
- Skate skiing: This technique is similar to ice skating, where the skier pushes off one ski at an angle and glides on the other. It requires a smooth, groomed surface and is typically faster than the classic technique.
Cross-country ski equipment consists of lightweight skis, boots, bindings, and poles. The boots are more flexible than those used in downhill skiing, allowing for better ankle movement and control.
2. Downhill Skiing:
Also known as alpine skiing, downhill skiing involves descending slopes at high speeds using skis and poles for balance and control. This sport is more focused on speed, technique, and navigating challenging terrain, including steep slopes, moguls, and even jumps.
Downhill skiing can be further categorized into several disciplines, such as slalom, giant slalom, super-G, and downhill racing. Each discipline has its own set of rules, courses, and ski equipment.
Downhill ski equipment includes heavier and stiffer skis, boots, bindings, and poles than those used in cross-country skiing. The boots are more rigid to provide better support and control during high-speed descents and sharp turns.
In summary, cross-country skiing is an endurance-based sport that involves traveling across flat or gently sloping terrain, while downhill skiing is focused on speed and technique as skiers navigate steeper slopes and challenging terrain. Both sports require specialized equipment and techniques, but they offer different experiences and challenges to participants.

Table 7: Conversation Prompt 2

"role": "system"
You are an AI visual assistant that can analyze a single image. You receive five sentences, each describing the same image you are observing. In addition, specific object locations within the image are given, along with detailed coordinates. These coordinates are in the form of bounding boxes, represented as (x1, y1, x2, y2) with floating numbers ranging from 0 to 1. These values correspond to the top left x, top left y, bottom right x, and bottom right y.
Using the provided caption and bounding box information, describe the scene in a detailed manner.
Instead of directly mentioning the bounding box coordinates, utilize this data to explain the scene using natural language. Include details like object counts, position of the objects, relative position between the objects.
When using the information from the caption and coordinates, directly explain the scene, and do not mention that the information source is the caption or the bounding box. Always answer as if you are directly looking at the image.

"role": "user"(5 captions+Bbox) (few-shot样例1)
A harbor filled with lots of boats next to a building.
A bicycle parked in front of several boats at a dock.
A red bicycle in front of a line of docked white yachts.
A bike sits before boats which sit before a long building.
A bicycle is a convenient means of land transportation when you live on a boat.
bicycle: [0.287, 0.641, 0.507, 0.874]
bicycle: [0.566, 0.667, 0.63, 0.731]
boat: [0.318, 0.579, 0.575, 0.724]
boat: [0.704, 0.607, 0.818, 0.727]
boat: [0.818, 0.601, 0.942, 0.744]
boat: [0.002, 0.53, 0.243, 0.71]
boat: [0.541, 0.611, 0.668, 0.731]
person: [0.778, 0.527, 0.797, 0.57]
cup: [0.708, 0.733, 0.724, 0.758]
boat: [0.236, 0.532, 0.404, 0.64]
boat: [0.81, 0.632, 0.836, 0.676]
boat: [0.957, 0.526, 1.0, 0.752]
"role": "assistant"
It is a harbor filled with numerous boats of various sizes docked next to a long building. Among the boats, there are a few white yachts lined up, standing out from the rest. There is a red bicycle prominently parked in front of the line of docked boats, serving as a convenient means of land transportation for those living on the boats. Another bicycle can be seen further back in the scene, near the middle of the harbor.
A person is visible near the right side of the harbor, possibly enjoying the view or attending to their boat. Additionally, there is a cup placed on a surface near the middle of the scene.

(省略两组few-shot样例...)
"role": "user"(5 captions+Bbox) (few-shot样例2)
"role": "assistant"
"role": "user"(5 captions+Bbox) (few-shot样例3)
"role": "assistant"

"role": "user"(5 captions+Bbox) (输入)
Describe the following image in detail:
A donut shop is full of different flavors of donuts. Fruit flavored donuts lined up in a glass fronted cabinet. A rack with some doughnuts in a glass case. A display case in a bakery filled with donuts. An assortment of doughnuts are arranged in a display case.
donut: [0.370, 0.584, 0.504, 0.709]; donut: [0.369, 0.220, 0.492, 0.317]; donut: [0.471, 0.587, 0.639, 0.706]; donut: [0.544, 0.213, 0.679, 0.316]; donut: [0.035, 0.220, 0.196, 0.328]; donut: [0.054, 0.608, 0.221, 0.711]; donut: [0.283, 0.586, 0.429, 0.708]; donut: [0.466, 0.226, 0.585, 0.320]; donut: [0.280, 0.232, 0.393, 0.322]; donut: [0.000, 0.609, 0.097, 0.722]

Table 8: Detail Prompt 2

A.2 Detail description & Reasoning Prompt

(system + 3×(user → assistant) + 1×user(目标图像); system_message.txt中定义了: 角色、任务、约束, user输入由目标图像的 5 captions 与 boxes组成。)

"role": "system"
You are an AI visual assistant that can analyze a single image. You receive five sentences, each describing the same image you are observing. In addition, specific object locations within the image are given, along with detailed coordinates. These coordinates are in the form of bounding boxes, represented as (x1, y1, x2, y2) with floating numbers ranging from 0 to 1. These values correspond to the top left x, top left y, bottom right x, and bottom right y.
The task is to use the provided caption and bounding box information, create a plausible question about the image, and provide the answer in detail.
Create complex questions beyond describing the scene.
To answer such questions, one should require first understanding the visual content, then based on the background knowledge or reasoning, either explain why the things are happening that way, or provide guides and help to user's request. Make the question challenging by not including the visual content details in the question so that the user needs to reason about that first.
Instead of directly mentioning the bounding box coordinates, utilize this data to explain the scene using natural language. Include details like object counts, position of the objects, relative position between the objects.
When using the information from the caption and coordinates, directly explain the scene, and do not mention that the information source is the caption or the bounding box. Always answer as if you are directly looking at the image.

"role": "user"(5 captions) (few-shot样例1)
A man wearing multiple neck ties making a goofy face.
A man in a white shirt wearing very many ties.
a man with ties on poses for a picture.
A man wearing multiple ties on his neck.
A young man smiles while wearing several ties. tie: [0.574, 0.298, 0.752, 0.704]; tie: [0.464, 0.339, 0.639, 0.789]; tie: [0.349, 0.363, 0.563, 0.732]; tie: [0.259, 0.255, 0.668, 0.805]; person: [0.019, 0.065, 0.962, 0.988]; person: [0.0, 0.24, 0.214, 1.0]; tie: [0.316, 0.778, 0.443, 0.867]; tie: [0.386, 0.707, 0.496, 0.801]; tie: [0.251, 0.354, 0.402, 0.747]; tie: [0.44, 0.318, 0.63, 0.397]; tie: [0.566, 0.33, 0.655, 0.543]; tie: [0.25, 0.409, 0.359, 0.556]
"role": "assistant"
Question:
What is unusual about this photo?
======
Answer:
In the photo, the man is wearing a total of ten ties around his neck. This is considered unusual because it is not common for someone to wear multiple ties at once. In a typical setting, a person would wear only one tie. The man's goofy facial expression also adds to the unconventional and humorous nature of the image.

(省略两组few-shot样例...)
"role": "user"(5 captions+Bbox) (few-shot样例2)
"role": "assistant"
"role": "user"(5 captions+Bbox) (few-shot样例3)
"role": "assistant"

"role": "user"(5 captions+Bbox) (输入)
Describe the following image in detail:
A newspaper that has sunglasses on top of it sitting in front of books. an apple sunglasses books and a teddy bear. A folded newspaper and sunglasses are on a table with an apple, books, and teddy bear behind. An apple sitting on a table next to sunglasses and a news paper. There are sunglasses laying on the folded newspaper.
tie: [0.258, 0.074, 0.527, 0.589]
apple: [0.621, 0.482, 0.853, 0.645]
book: [0.154,0.107, 0.275, 0.590]
book: [0.535, 0.090, 0.735, 0.583]
teddy bear: [0.753, 0.084, 1.000, 0.517]
book: [0.681, 0.097, 0.796, 0.483]
book: [0.443, 0.099, 0.574, 0.588]
book:[0.267, 0.337, 0.386, 0.579]

Table 9: Detail Prompt 2